A Detailed methodology

Here are some additional details to the methodology sections.

A.1 Methodology relating to all experiments

Truncation All samples were truncated to 4096 tokens.

Model details The model's dtype was bfloat16.

Datasets All datasets were initially randomly shuffled with seed 13.

Text generation All generated text was produced without sampling.

A.2 Broad steering experiments

We used the following injection coefficients:

$$0.0, 0.25, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8,$$
$$1.9, 2.0, 2.25, 2.5, 2.75, 3, 3.5, 4, 4.5, 5, 6, 7, 8, 9, 10, 15, 20, 30, 40, 50$$

There are some criteria for which we did not stepwise go through these values. We calculate the steered top-1 accuracy score divided by the default score. Based on this relative score, we used the following strategy to go through the injection coefficients. The score of either general or Python code was below 0.05, the run was stopped. Otherwise, if the relative score was below 0.15, a step size of 5 would be taken instead of 1.

A.3 Multi-steering experiments

Grid search Extending on the grid search described in Section 2.2. The tested injections coefficients values were $\{0.5, 1, 2, 3, 5, 10, 20, 30, 40, 60, 80, 120, 200, 300\}$, and the tested layers were $\{0, 5, 10, 15, 20, 25, 29, 31\}$. The specific injection coefficients per behaviour are shown in Table 1 for layers 10 and 15, the two most effective layers.

	Agreeableness	Anti Immigration	Myopic	Wealth seeking	Sycophancy
Layer 10	0.5, -3	3, -1	10, -1	1, -2	1, -20
Layer 15	0.5, -1	1, -0.5	2, -1	1, -2	2, -5

Table 1: The injection coefficients for each concept after performing grid search for adding and subtracting the steering vectors for layers 10 and 15.

B Layer 10 results

B.1 Single steering

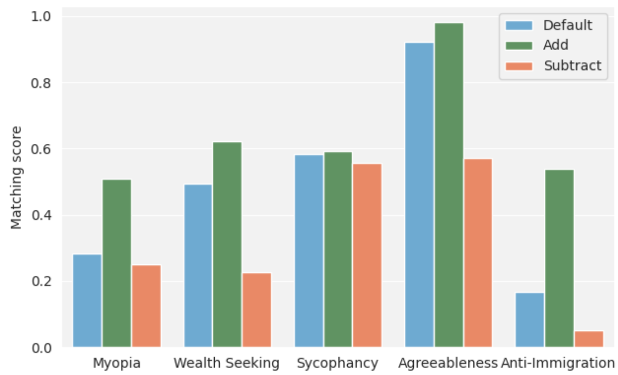

Figure 8: The results for individual steering for layer 10. The used injection coefficients can be found in Table 1

B.2 Combined steering

Figure 9: Combining the individual steering vectors into one injected in layer 10. The combinations differ in three dimensions: take the mean or sum, weighted or unweighted, and subtracted or added. We compare the combined steering to the individual steering presented in Figure 8, which are indicated with the grey horizontal lines.

C Activation distribution

C.1 General coding activation distributions

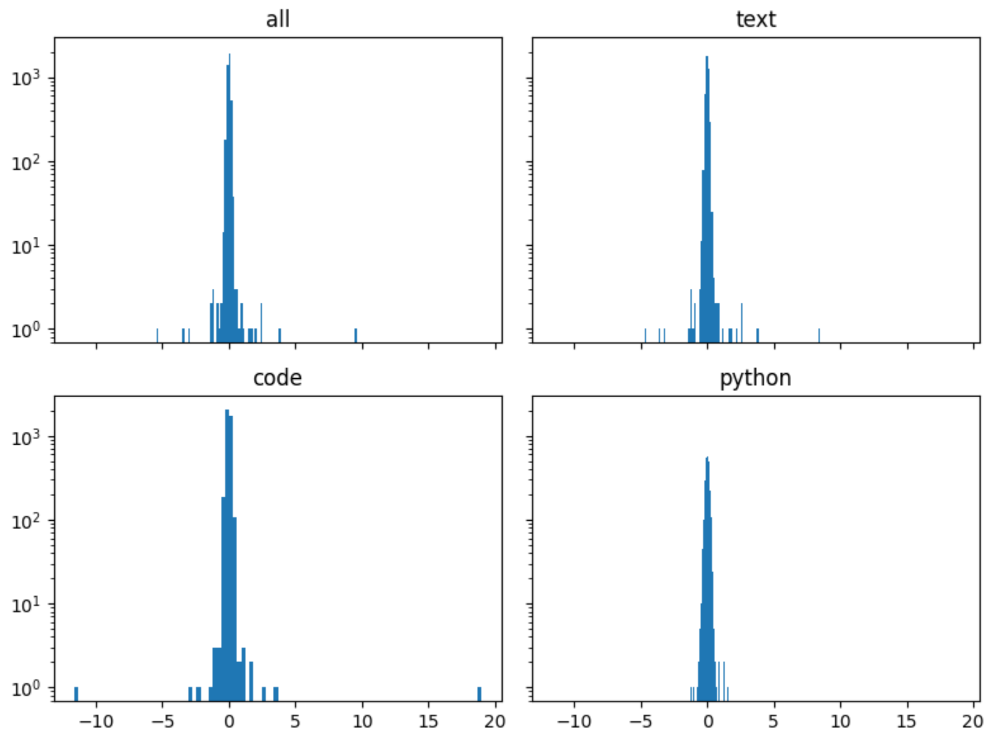

Figure 10: The distribution of activations for layer 15 given multiple datasets.

C.2 Multi steering activation distributions

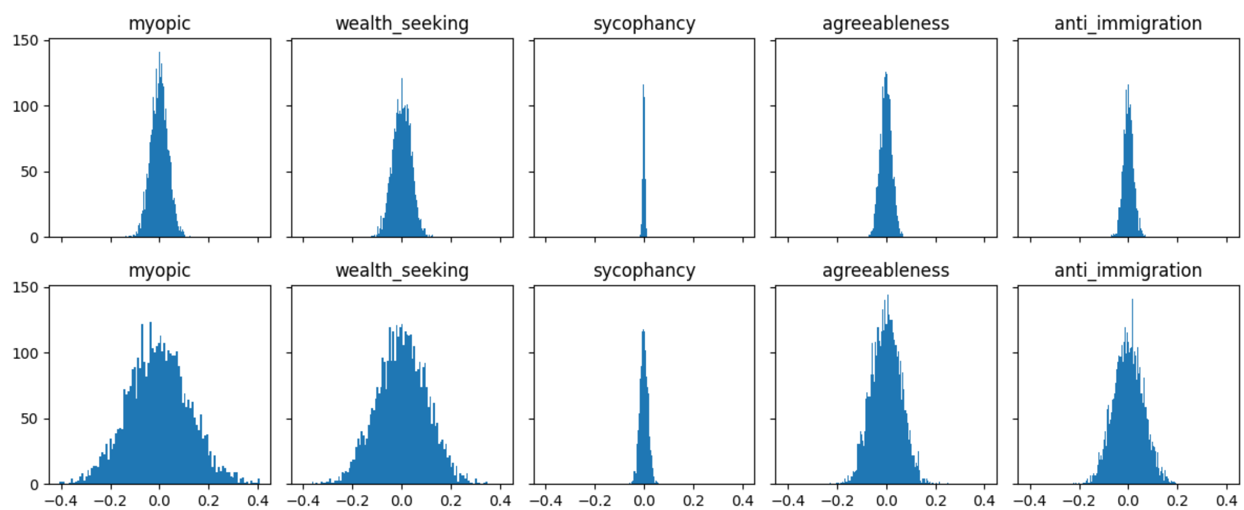

Figure 11: The distribution of activations for layers 10 (above) and 15 (below) for numerous concepts.